

JEE MAIN REVISION MASTER: IUPAC NOMENCLATURES 2026 1 QUESTIONS 1.67% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 5 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Kcet Iupac Nomenclature - Core Application	5 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Iupac Nomenclature - Core Application (Moderate Yield — 5 PYQs)

Core Principle: Key problem pattern in Kcet Iupac Nomenclature.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The compound with molecular formula $C_{(20)}H_{(42)}$ is..." [Ans: C]
- [2025 Shift] Q#2 (MCQ): "The IUPAC name of the given organic compound is $HC \equiv C-CH=CH-CH=CH_{(2)}...$ " [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "IUPAC name of the compound is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The compound with molecular formula $C_{(20)}H_{(42)}$ is

- (A) Decane
- (B) Dodecane
- (C) Eicosane [Correct]
- (D) Hicosane

Correct Answer: C

Solution Analysis: The molecular formula is $C_{(20)}H_{(42)}$. For alkanes, the general formula is $C_{(n)}H_{(2n+2)}$. Here, $n=20$. So, $H=2(20)+2=42$. Hence the compound is an alkane with 20 carbon atoms. The alkane with 20 carbon atoms is called Eicosane. So, the correct option is Option C: Eicosane.